BISO JICA

# MODEL FOR REVITALIZING DEPLETED APRICOT TREE SOIL – COMPREHENSIVE RESTORATION THROUGH MICROBIAL ENZYMES

## CULTIVATION GOALS

- Enhances compromised and nutrient-poor soil; reinstates
- beneficial microorganisms; improves soil aeration and
- revitalizes plant growth potential.

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### PRODUCTS FOR APPLICATION

- Daichi No Tomo (microbial enzyme) Daichi
- No Tomo 500 (concentrated formulation)

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## IMPLEMENTATION STRATEGY

Preliminary stage:

- Location: Phuong Binh Apricot Blossom Garden (a garden recognized for its deteriorating condition and significant number of dead trees) Area: 1 hectare
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Initial condition:

- The apricot trees are fragile, with underdeveloped roots.
- Each year, 20-30% of the trees perish.
- Fertilizers and pesticides cost 130 million VND annually.
- Labor consists of five individuals.
- Electricity and water expenses amount to approximately 12 million VND per month.

Applying the Biso Jica organic method: Organic-chemical ratio: 70% - 30%

- Spray uniformly across the entire surface of the soil (for both potted plants and garden beds).
- Combine 100ml of Daichi No Tomo with 1L of Daichi No Tomo 500 and dilute with 500L of water.
- 10,000 $m^2$ – (2000 liters of water) retail price 500 * 4 = 2 million.
- Dilute in water to irrigate the entire 10,000 $m^2$ area.

Combine SP 10-15-20 with vermicompost or another organic fertilizer at a ratio of 30%.

Learning

Outcomes:

- After 15 days: The plant produces new shoots, revitalizes its root system, and exhibits healthy growth.
- After 30 days: The leaves regenerate uniformly, and the plant reaches full development.

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**REGULAR IRRIGATION (BEGINNING IN THE SECOND MONTH):**

- Frequency: 1–2 times per month
- Dosage: 100 ml Daichi No Tomo + 1 L Daichi No Tomo 500 diluted in 500–700 L of water
- Maintenance cost: 2–4 million VND per month

Effective:

The plant features large, uniformly sized leaves that contribute to the reduction of pests and diseases, leading to a naturally robust plant with a resilient root system.

The branches and leaves grow uniformly and thoroughly, as evidenced by clips from the Saigon Market channel, particularly in comparison to extensive apricot blossom gardens.

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**LAND MANAGEMENT POST-RENOVATION**

- Refrain from using chemical fertilizers during the enzyme treatment phase (for at least the initial 2-3 months) if the trees are significantly weakened or afflicted by diseases, such as stem cracking and sap leakage in durian trees. It
- is advisable to integrate composted straw, well-rotted manure, coffee husks, and burnt rice husks with Nioi Zero + BK Partner enzymes and Biso fermentation starter to enhance efficacy while reducing costs.

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## COST: Implementation commencement period: 7 months

- 25 million VND allocated for BISO products, 10 million VND for supplementary chemical fertilizers.
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Compared to the previous 130 million per year, savings:

- Labor costs, electricity, and water expenses have been decreased by 50%, and the application of pesticides and chemical fertilizers has been markedly diminished.

Result:

- Loss rate < 5% (Preserving numerous nearly worthless trees)
- Healthy trees exhibit stable growth and are less vulnerable to pests and diseases, presenting a distinct appearance compared to orchards employing inorganic farming methods.

In comparison to other nurseries, the trees exhibit uneven growth, with many developing slowly and remaining weak, leading to a loss rate of 15-20%. A variety of formulas and products must be utilized in combination. Numerous pests and diseases are present, and they demonstrate resistance to pesticides, particularly when contrasted with the renowned, aesthetically pleasing male apricot blossom garden.

## PROFESSIONAL OBSERVATIONS

- It is recommended to undertake renovations prior to the onset of the rainy season or the commencement of the dry season to effectively manage humidity levels.
  Planting prior to the onset of the rainy season or the commencement of the dry season facilitates the management of moisture levels, as the grower's technique plays a crucial role in the cultivation process.
- Growers must diligently monitor water conditions, climate, and soil
- responses. Maintaining a spraying and irrigation log is advisable, noting the time, humidity, climate, soil reactions, and weed presence.
- The improvement area can be categorized into soil groups A, B, and C to
- evaluate effectiveness. Additional enzyme strains may be incorporated to enhance foliage protection: Mikata 1000 / Mikata 1200.